

Name: Santiago, Jalene	Date Performed: March 12, 2026
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Instructor: Engr. Robin Valenzuela	Semester and SY: 2nd Sem - 25-26

Midterm Skills Exam: Install, Configure, and Manage Log Monitoring tools

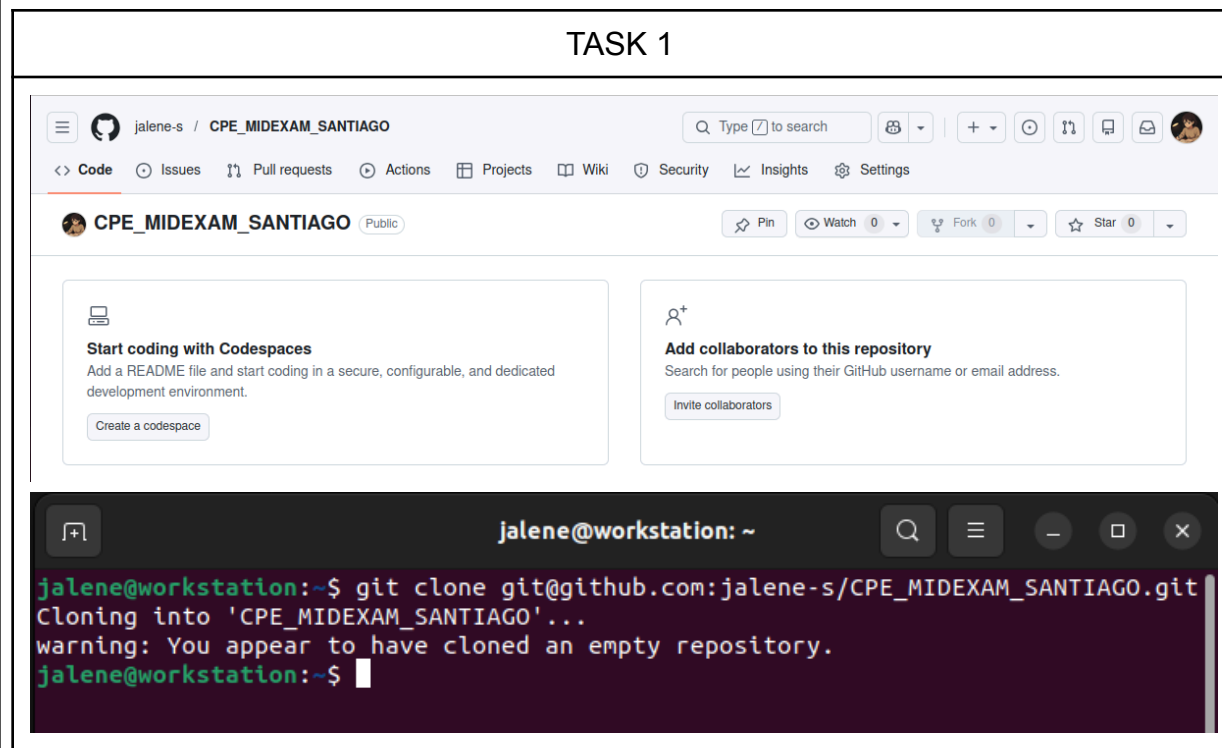
1. Objectives

Create and design a workflow that installs, configure and manage enterprise availability, performance and log monitoring tools using Ansible as an Infrastructure as Code (IaC) tool.

2. Instructions

1. Create a repository in your GitHub account and label it CPE_MIDEXAM_SURNAME.

TASK 1



The screenshot displays a GitHub repository page for 'jalene-s / CPE_MIDEXAM_SANTIAGO'. The repository is public and has 0 watches, 0 forks, and 0 stars. Below the repository name, there are two main sections: 'Start coding with Codespaces' and 'Add collaborators to this repository'. The 'Start coding with Codespaces' section includes a button to 'Create a codespace'. The 'Add collaborators to this repository' section includes a button to 'Invite collaborators'. Below these sections, there is a terminal window showing the command to clone the repository: `git clone git@github.com:jalene-s/CPE_MIDEXAM_SANTIAGO.git`. The terminal output shows the repository being cloned into 'CPE_MIDEXAM_SANTIAGO' and a warning message: 'warning: You appear to have cloned an empty repository.'

2. Clone the repository and do the following:
 - 2.1. Create an Ansible playbook that does the following with an input of a config.yaml file and arranged Inventory file:
 - 2.2. Install and configure Elastic Stack in separate hosts (Elastic Search, Kibana, Logstash) • Install Nagios in one host

TASK 2.2

Ubuntu - Elastic Stack

```
! config.yaml u ! main.yml elastic-stack_ubuntu/... U X ! main.yml lamp-stack_centos/... U ! main.yml elastic-stack_centos/... U
elastic-stack_ubuntu > tasks > ! main.yml
1 ---
2 - name: Add Elastic GPG key
3   apt_key:
4     url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
5     state: present
6
7 - name: Add Elastic repository
8   apt_repository:
9     repo: "deb https://artifacts.elastic.co/packages/8.x/apt stable main"
10    state: present
11    filename: elastic
12
13 - name: Update package cache
14   apt:
15     update_cache: yes
16
17 - name: Install Elasticsearch
18   apt:
19     name: elasticsearch
20     state: present
21
22 - name: Install Logstash
23   apt:
```

```
! config.yaml u ! main.yml elastic-stack_ubuntu/... U X ! main.yml lamp-stack_centos/... U ! main.yml elastic-stack_centos/... U
elastic-stack_ubuntu > tasks > ! main.yml
21
22 - name: Install Logstash
23   apt:
24     name: logstash
25     state: present
26
27 - name: Install Kibana
28   apt:
29     name: kibana
30     state: present
31
32 - name: Start and enable Elasticsearch
33   systemd:
34     name: elasticsearch
35     enabled: yes
36     state: started
37
38 - name: Start and enable Logstash
39   systemd:
40     name: logstash
41     enabled: yes
42     state: started
43
```

```
44 - name: Start and enable Kibana
45   systemd:
46     name: kibana
47     enabled: yes
48     state: started
```

Ubuntu - Nagios

```

! elastic-stack_centos/... U ! main.yml lamp-stack_ubuntu/... U ! main.yml nagios/tasks U ! main.yml nagios_pre-requisite/... U X
nagios_pre-requisite > tasks > ! main.yml
1 ---
2 - name: Install Nagios Pre-Requisites on Ubuntu
3   apt:
4     name:
5       - apache2
6       - php
7       - php-gd
8       - libapache2-mod-php
9       - unzip
10      - wget
11      - build-essential
12      - nagios-plugins
13     state: present
14     update_cache: yes
15   when: ansible_distribution == "Ubuntu"
16
17 - name: Enable Apache service
18   service:
19     name: apache2
20     state: started
21     enabled: yes
22   when: ansible_distribution == "Ubuntu"
23

```

```

! main.yml elastic-stack_centos/... U ! main.yml lamp-stack_ubuntu/... U ! main.yml nagios/tasks U X
nagios > tasks > ! main.yml
1 ---
2 - name: Install Nagios on Ubuntu
3   apt:
4     name: nagios4
5     state: present
6     update_cache: yes
7   when: ansible_distribution == "Ubuntu"
8
9 - name: Start Nagios Service on Ubuntu
10  service:
11    name: nagios4
12    state: started
13    enabled: yes
14  when: ansible_distribution == "Ubuntu"

```

CentOS - Elastic Stack

```

! main.yml elastic-stack_centos/... U X ! main.yml lamp-stack_ubuntu/... U ! main.yml nagios/tasks U ! main.yml nagios_pre-requisite/... U X
elastic-stack_centos > tasks > ! main.yml
1 ---
2 - name: Add Elastic GPG key
3   rpm_key:
4     key: https://artifacts.elastic.co/GPG-KEY-elasticsearch
5     state: present
6
7 - name: Add Elastic repository
8   dnf:
9     name: elastic-8.x
10    description: Elastic repository for 8.x packages
11    baseurl: https://artifacts.elastic.co/packages/8.x/yum
12    gpgcheck: yes
13    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
14    enabled: yes
15
16 - name: Install Elasticsearch
17   dnf:
18     name: elasticsearch
19     state: present
20
21 - name: Install Logstash
22   dnf:
23     name: logstash

```

```
! main.yml elastic-stack_centos/... U X ! main.yml lamp-stack_ubuntu/... U ! main.yml nagios/tasks U ! main.yml nagios_pre-requisit...
elastic-stack_centos > tasks > ! main.yml
20
21 - name: Install Logstash
22   dnf:
23     name: logstash
24     state: present
25
26 - name: Install Kibana
27   dnf:
28     name: kibana
29     state: present
30
31 - name: Start and enable Elasticsearch
32   systemd:
33     name: elasticsearch
34     enabled: yes
35     state: started
36
37 - name: Start and enable Logstash
38   systemd:
39     name: logstash
40     enabled: yes
41     state: started
42
43 - name: Start and enable Kibana
44   systemd:
45     name: kibana
46     enabled: yes
47     state: started
```

2.3. Install Lamp Stack in separate hosts (Httpd + Php,Mariadb)

TASK 2.3

Ubuntu - Lamp Stack

```
! main.yml elastic-stack_centos/... U ! main.yml lamp-stack_ubuntu/... U X ! main.yml nagios/tasks U ! main.yml nagios_pre-requisit...
lamp-stack_ubuntu > tasks > ! main.yml
1 ---
2 - name: Install Apache, MySQL, and PHP
3   apt:
4     name: "{{ item }}"
5     state: present
6   loop:
7     - apache2
8     - mysql-server
9     - php
10    - libapache2-mod-php
11    - php-mysql
12
13 - name: Start and enable Apache
14   systemd:
15     name: apache2
16     enabled: yes
17     state: started
18
19 - name: Start and enable MySQL
20   systemd:
21     name: mysql
22     enabled: yes
23     state: started
```

```
! main.yml lamp-stack_centos/... U X ! main.yml elastic-stack_centos/... U ! main.yml lamp-stack_ubuntu/... U ! main.yml nagios/tasl ...
lamp-stack_centos > tasks > ! main.yml
1 ---
2 - name: Install Apache, MySQL, and PHP
3   dnf:
4     name: "{{ item }}"
5     state: present
6   loop:
7     - httpd
8     - mysql-community-server
9     - php
10    - php-mysql
11
12 - name: Start and enable Apache
13   systemd:
14     name: httpd
15     enabled: yes
16     state: started
17
18 - name: Start and enable MySQL
19   systemd:
20     name: mysql
21     enabled: yes
22     state: started
```

3. Document all your tasks using this document. Provide proofs of all the ansible playbooks codes and successful installations.
4. Document the push and commit from the local repository to GitHub.
5. Finally, paste also the link of your GitHub repository in the documentation.

3. Output (screenshots and explanations)

config.yml

```
! config.yml U X ! main.yml elastic-stack_ubuntu/... U ! main.yml lamp-stack_centos/... U
! config.yml
1 ---
2 - name: Ubuntu (Elastic Stack, Lamp Stack, Nagios)
3   hosts: ubuntu
4   become: true
5
6   roles:
7     - pre-requisites
8     - elastic-stack_ubuntu
9     - lamp-stack_ubuntu
10    - nagios_pre-requisite
11    - nagios
12
13 - name: CentOS (Elastic Stack and Lamp Stack)
14   hosts: centos
15   become: true
16
17   roles:
18     - pre-requisites
19     - elastic-stack_centos
20    - lamp-stack_centos
```

This is the main playbook that will perform all the tasks I configured using roles. Each time it runs, it will call all the roles and perform all the tasks. I divided the tasks into 2 (Ubuntu and CentOS) to maintain organization and clean

code. For Ubuntu, the roles were elastic and lamp stack and also Nagios. For CentOS, the roles were elastic and lamp stack. In both servers, I made sure that I use the correct package names and commands to implement all the tasks.

Elastic Stack Output (Ubuntu and CentOS)

Manage Node 1
(server1) Ubuntu

```
TASK [pre-requisites : Install Pre-requisites in Ubuntu] ***
ok: [server1]
```

```
TASK [elastic-stack_ubuntu : Add Elastic GPG key] ***
ok: [server1]
```

```
TASK [elastic-stack_ubuntu : Add Elastic repository] ***
ok: [server1]
```

```
TASK [elastic-stack_ubuntu : Update package cache] ***
changed: [server1]
```

```
TASK [elastic-stack_ubuntu : Install Elasticsearch] *****
ok: [server1]
```

```
TASK [elastic-stack_ubuntu : Install Logstash] *****
ok: [server1]
```

```
TASK [elastic-stack_ubuntu : Install Kibana] *****
changed: [server1]
```

```
TASK [elastic-stack_ubuntu : Start and enable Elasticsearch]
changed: [server1]
```

```
TASK [elastic-stack_ubuntu : Start and enable Logstash] *****
changed: [server1]
```

```
TASK [elastic-stack_ubuntu : Start and enable Kibana] *****
changed: [server1]
```

```
jalene@server1:~$ sudo systemctl status elasticsearch
[sudo] password for jalene:
● elasticsearch.service - Elasticsearch
   Loaded: loaded (/lib/systemd/system/elasticsearch.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2026-03-12 18:21:45 +08; 30min ago
     Main PID: 10000 (java)
   CGroup: /systemd/system/elasticsearch.service
```

```
jalene@server1:~$ sudo systemctl status kibana
● kibana.service - Kibana
   Loaded: loaded (/lib/systemd/system/kibana.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2026-03-12 18:22:11 +08; 30min ago
     Main PID: 10000 (java)
   CGroup: /systemd/system/kibana.service
```

```
jalene@server1:~$ sudo systemctl status logstash
● logstash.service - logstash
   Loaded: loaded (/lib/systemd/system/logstash.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2026-03-12 18:51:23 +08; 1min 46s ago
     Main PID: 10000 (java)
   CGroup: /systemd/system/logstash.service
```

Manage Node 2 (server4) CentOS

```
TASK [pre-requisites : Install Pre-requisites in CentOS] **  
changed: [server4]
```

```
TASK [elastic-stack_centos : Add Elastic GPG key] *****  
changed: [server4]
```

```
TASK [elastic-stack_centos : Add Elastic repository] ***  
ok: [server4]
```

```
TASK [elastic-stack_centos : Install Elasticsearch] *****  
changed: [server4]
```

```
TASK [elastic-stack_centos : Install Logstash] *****  
changed: [server4]
```

```
TASK [elastic-stack_centos : Install Kibana] *  
changed: [server4]
```

The task was to configure and install elastic stack (elastic search, kibana, and logstash) in both Ubuntu and CentOS servers. The output shows that most installations were successful.

Nagios (Ubuntu)

Manage Node 1 (server1) Ubuntu

```
TASK [nagios_pre-requisite : Install Nagios Pre-Requisites on Ubuntu]  
ok: [server1]
```

```
TASK [nagios_pre-requisite : Enable Apache service] *****  
ok: [server1]
```

```
TASK [nagios_pre-requisite : Enable Apache modules needed for Nagios]  
changed: [server1]
```

```
TASK [nagios_pre-requisite : Restart Apache service] *****  
changed: [server1]
```

```
TASK [nagios : Install Nagios on Ubuntu] *****  
ok: [server1]
```

```
TASK [nagios : Start Nagios Service on Ubuntu] *****  
ok: [server1]
```

```
jalene@server1:~$ sudo systemctl status nagios  
Unit nagios.service could not be found.  
jalene@server1:~$ sudo systemctl status nagios4  
● nagios4.service - nagios4  
   Loaded: loaded (/lib/systemd/system/nagios4.service; enabled; vendor prese  
   Active: active (running) since Thu 2026-03-12 15:00:18 +08; 3h 53min ago  
     Docs: man:nagios4  
  Main PID: 641 (nagios4)  
    Tasks: 6 (limit: 1054)  
   Memory: 1.4M  
      CPU: 26.500s  
   CGroup: /system.slice/nagios4.service  
           └─641 /usr/sbin/nagios4 /etc/nagios4/nagios.cfg  
             └─650 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh  
               └─651 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh  
                 └─652 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh  
                   └─653 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh  
                     └─719 /usr/sbin/nagios4 /etc/nagios4/nagios.cfg
```

Nagios was only installed on the Ubuntu server. Before installing and starting Nagios, I have to ensure first that all dependencies for it are installed. This output shows that Nagios is successfully installed and started.

Lamp Stack Output (Ubuntu and CentOS)

Manage Node 1 (server1) Ubuntu	<pre>TASK [lamp-stack_ubuntu : Install Apache, MySQL, and PHP] * ok: [server1] => (item=apache2) changed: [server1] => (item=mysql-server) ok: [server1] => (item=php) ok: [server1] => (item=libapache2-mod-php) changed: [server1] => (item=php-mysql) TASK [lamp-stack_ubuntu : Start and enable Apache] ***** ok: [server1]</pre>
Manage Node 2 (server4) CentOS	

The Lamp Stack (httpd (apache) + php and mariadb) were also installed on both Ubuntu and CentOS. Due to time constraints, I wasn't able to accomplish all tasks.

GITHUB PUSH

```
jalene@workstation:~/CPE_MIDEXAM_SANTIAGO$ git commit -m "Midterm Exam"
create mode 100644 nagios/meta/main.yml
create mode 100644 nagios/tasks/main.yml
create mode 100644 nagios/tests/inventory
create mode 100644 nagios/tests/test.yml
create mode 100644 nagios/vars/main.yml
create mode 100644 nagios_pre-requisite/README.md
create mode 100644 nagios_pre-requisite/defaults/main.yml
create mode 100644 nagios_pre-requisite/handlers/main.yml
create mode 100644 nagios_pre-requisite/meta/main.yml
create mode 100644 nagios_pre-requisite/tasks/main.yml
create mode 100644 nagios_pre-requisite/tests/inventory
create mode 100644 nagios_pre-requisite/tests/test.yml
create mode 100644 nagios_pre-requisite/vars/main.yml
create mode 100644 pre-requisites/README.md
create mode 100644 pre-requisites/defaults/main.yml
create mode 100644 pre-requisites/handlers/main.yml
create mode 100644 pre-requisites/meta/main.yml
create mode 100644 pre-requisites/tasks/main.yml
create mode 100644 pre-requisites/tests/inventory
create mode 100644 pre-requisites/tests/test.yml
create mode 100644 pre-requisites/vars/main.yml
• jalene@workstation:~/CPE_MIDEXAM_SANTIAGO$ git push
Enumerating objects: 87, done.
Counting objects: 100% (87/87), done.
Delta compression using up to 3 threads
Compressing objects: 100% (36/36), done.
Writing objects: 100% (87/87), 8.64 KiB | 680.00 KiB/s, done.
Total 87 (delta 6), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (6/6), done.
To github.com:jalene-s/CPE_MIDEXAM_SANTIAGO.git
 * [new branch]    main -> main
• jalene@workstation:~/CPE_MIDEXAM_SANTIAGO$
```

GitHub link:

https://github.com/jalene-s/CPE_MIDEXAM_SANTIAGO

Conclusion:

Our skills exam is focused on designing a workflow that installs, configures and manages enterprise availability, performance and log monitoring tools using Ansible as an Infrastructure as Code (IaC) tool. For instance, we installed two different stacks (Elastic and Lamp), both containing multiple packages. During the exam, I encountered difficulties as not all packages are familiar to me and I wasn't sure at all on how to run them. However, I somehow accomplished most of the tasks and was able to use my knowledge in Ansible for this exam.